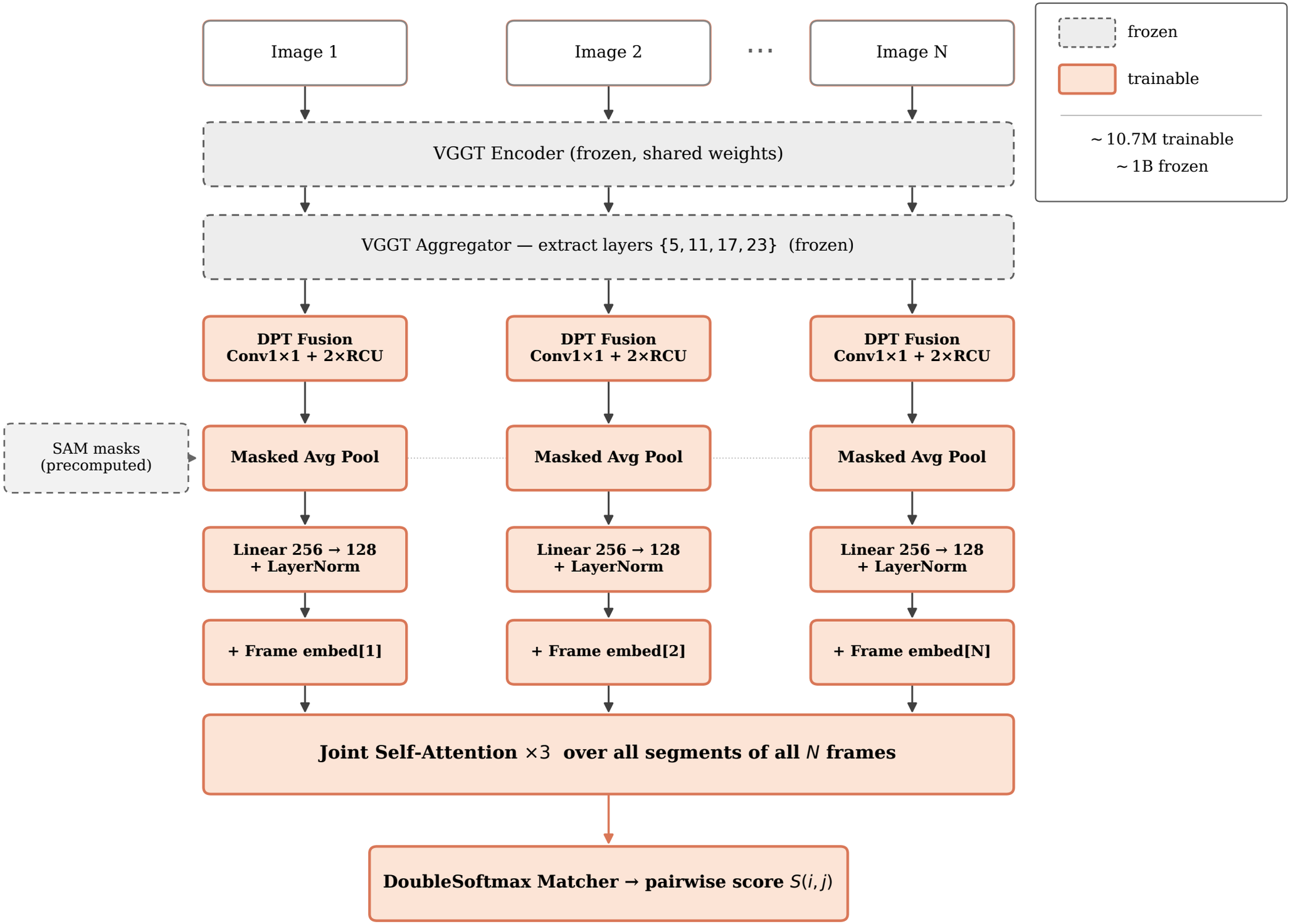
Image 1
Image 2
···
Image N
frozen
trainable
~10.7M trainable
~1B frozen
VGGT Encoder (frozen, shared weights)
VGGT Aggregator — extract layers {5, 11, 17, 23} (frozen)
DPT Fusion
Conv1×1 + 2×RCU
DPT Fusion
Conv1×1 + 2×RCU
DPT Fusion
Conv1×1 + 2×RCU
SAM masks
(precomputed)
Masked Avg Pool
Masked Avg Pool
Masked Avg Pool
Linear 256 → 128
+ LayerNorm
Linear 256 → 128
+ LayerNorm
Linear 256 → 128
+ LayerNorm
+ Frame embed[1]
+ Frame embed[2]
+ Frame embed[N]
Joint Self-Attention ×3 over all segments of all N frames
DoubleSoftmax Matcher → pairwise score S(i, j)